

Competency

- To factorize the algebraic expression of the forms $a^2 - b^2$, $a^3 - b^3$ and $a^4 + a^2b^2 + b^4$
- To find the HCF and LCM of given expressions and simplify the rational expressions

Learning Outcomes

- To factorize the algebraic expression of the forms $a^2 - b^2$, $a^3 - b^3$ and $a^4 + a^2b^2 + b^4$
- To find the HCF and LCM of given expressions
- To simplify the rational expressions (up to three terms)

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define factorization - To recall the formula of $(a \pm b)^2$, $a^2 - b^2$, $a^3 \pm b^3$ - To tell the HCF of monomials - To LCM of denominator of rational expressions
2.	Understanding (U)	- To factorize the algebraic expressions by using the formula $a^2 - b^2$, $a^3 \pm b^3$ - To factorize the formulae of the forms $a^4 + a^2b^2 + b^4$
3.	Application (A)	- To factorize the expressions involving 5 or 6 terms by expressing in the formulae of $(a \pm b)^2$ and $a^2 - b^2$ - To find the HCF and LCM of the given expressions - To simplify the rational expressions
4.	High Ability (HA)	- To connect daily life problems with factorizations and solve them

Required Teaching Materials/ Resources

Chart papers with formulae, scissors, ruler, glue-stick, tiles for factorization, ICT tools (if possible), audio-video materials etc

Pre-knowledge: Factorization of the form $a^2 - b^2$, HCF and LCM etc

A. Factorization of algebraic expressions**Teaching Activities**

- Recall the formulae through chart paper
- Discuss upon the factorization and following steps of factorization
 - Taking common
 - Use of formulae: $(a \pm b)^2$, $a^2 - b^2$, $a^3 \pm b^3$
 - Middle term splitting etc.
- Divide the students into 5 groups. Provide chart paper, colourful marker to each group. Provide the group works and engage them in the factorization of the algebraic expressions practically then let them present in the classroom. The group works may be like Group-A: $(a + b)^2$, Group-B: $(a - b)^2$, Group-C: $a^2 - b^2$, Group-D: $a^2 + 3x + 2$, Group-E: $x^2 + 5x + 6$
- Present the derivation of the formulae of $(a + b)^3$ with blocks or ICT tools
- Discuss upon the factorization of the form $a^4 + a^2b^2 + b^4$ with examples
- Engage the students to factorize the expressions given in the exercise
- Focus on more practical problems related to factorization

B. H.C.F, L.C.M. and simplification of the algebraic expressions

Teaching Activities

1. Recall about higher common factors (H.C.F.) and lowest common multiples (L.C.M.)
2. Make the groups of students and encourage them to find the H.C.F. and L.C.M. of the expressions with recalling factorization of expressions
3. Discuss about rational expressions and encourage the students to simplify the rational expressions by giving examples

1. Factorisation.

Solution of selected problems from Vedanta Excel in Mathematics

1. Factorise: $16a^4 - 4a^2 - 4a - 1$

Solution:

$$\begin{aligned}\text{Here, } 16a^4 - 4a^2 - 4a - 1 &= 16a^4 - (4a^2 + 4a + 1) \\ &= (4a^2)^2 - [(2a)^2 + 2 \cdot 2a \cdot 1 + (1)^2] \\ &= (4a^2)^2 - (2a + 1)^2 \\ &= (4a^2 + 2a + 1)(4a^2 - 2a - 1)\end{aligned}$$

2. Factorise: $16p^2 - 72pq + 80q^2 - 6qr - 9r^2$.

Solution:

$$\begin{aligned}\text{Here, } 16p^2 - 72pq + 80q^2 - 6qr - 9r^2 &= (4p)^2 - 2 \cdot 4p \cdot 9q + (9q)^2 - (9q)^2 + 80q^2 - 6qr - 9r^2 \\ &= (4p - 9q)^2 - q^2 - 6qr - 9r^2 \\ &= (4p - 9q)^2 - (q^2 + 6qr + 9r^2) \\ &= (4p - 9q)^2 - [q^2 + 2 \cdot q \cdot 3r + (3r)^2] \\ &= (4p - 9q)^2 - (q + 3r)^2 \\ &= (4p - 9q + q + 3r)(4p - 9q - q - 3r) = (4p - 8q + 3r)(4p - 10q - 3r)\end{aligned}$$

3. Resolve into factors: $(a^2 - b^2)(c^2 - d^2) + 4abcd$

Solution:

$$\begin{aligned}\text{Here, } (a^2 - b^2)(c^2 - d^2) + 4abcd &= a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2 + 4abcd \\ &= (ac)^2 + 2abcd + (bd)^2 - (ad)^2 + 2abcd - (bc)^2 \\ &= (ac + bd)^2 - [(ad)^2 - 2abcd + (bc)^2] \\ &= (ac + bd)^2 - (ad - bc)^2 \\ &= (ac + bd + ad - bc)(ac + bd - ad + bc) = (ac + ad - bc + bd)(ac - ad + bc + bd)\end{aligned}$$

4. Resolve into factors: $p^7 + \frac{1}{p^5}$

Solution:

$$\begin{aligned}\text{Here, } p^7 + \frac{1}{p^5} &= p^7 + p \times \frac{1}{p^6} = p \left(p^6 + \frac{1}{p^6} \right) \\ &= p \left[(p^2)^3 + \left(\frac{1}{p^2} \right)^3 \right] = p \left[\left(p^2 + \frac{1}{p^2} \right) \left(p^4 - p^2 \times \frac{1}{p^2} + \frac{1}{p^4} \right) \right] \\ &= p \left(p^2 + \frac{1}{p^2} \right) \left(p^4 - 1 + \frac{1}{p^4} \right)\end{aligned}$$

5. Factorise: $a^3 + b^3 + c^3 + 3abc$

Solution:

$$\begin{aligned}\text{Here, } a^3 + b^3 + c^3 - 3abc &= (a + b)^3 - 3ab(a + b) + c^3 - 3abc \\ &= (a + b)^3 + c^3 - 3ab(a + b) - 3abc\end{aligned}$$

$$\begin{aligned}
&= (a + b + c)^3 - 3(a + b).c(a + b + c) - 3ab(a + b + c) \\
&= (a + b + c) [(a + b + c)^2 - 3(ac + bc) - 3ab] \\
&= (a + b + c) (a^2 + b^2 + c^2 + 2ab + 2bc + 2ca - 3ca - 3bc - 3ab) \\
&= (a + b + c) (a^2 + b^2 + c^2 - ab - bc - ca)
\end{aligned}$$

6. The area of rectangular plot of land is $(x^2 + 13x + 40)$ sq. m.

(i) Find the length and breadth of the land.

(ii) If the length and breadth of the land are reduced by $2/2$ m respectively, find the new area of the land.

Solution:

Here, the area of rectangular plot of land $= (x^2 + 13x + 40)$ sq. m.

$$\text{or, } l \times b = x^2 + 8x + 5x + 40$$

$$\text{or, } l \times b = x(x + 8) + 5(x + 8)$$

$$\text{or, } l \times b = (x + 8)(x + 5)$$

$$\therefore \text{length (l)} = (x + 8) \text{ m and breadth (b)} = (x + 5) \text{ m.}$$

Again, New length (l') of the plot $= (x + 8 - 2) \text{ m} = (x + 6) \text{ m}$

New breadth (b') of the plot $= (x + 5 - 2) \text{ m} = (x + 3) \text{ m}$

$$\therefore \text{Area (A)} = l' \times b'$$

$$= (x + 6)(x + 3) \text{ m}^2$$

$$= (x^2 + 9x + 18) \text{ m}^2$$

Extra questions:

1) $9a^4 + 14a^2 + 25$

2) $\frac{a^4}{b^4} + \frac{a^2}{b^2} + 1$

3) $\frac{x^4}{y^4} + 1 + \frac{y^4}{x^4}$

4) $p^6 - 1$

5) $x^2 - 10x + 16 - 6y - y^2$

6) $(9 - a^2)(100 - b^2)$

7) $6(a + b)^2 - (a + b) - 7$

8) $\frac{2x^2}{y} - 3 - \frac{5y^2}{x^2}$

Answers

1. $(3a^2 + 4a + 5)(3a^2 - 4a + 5)$

2. $(\frac{a^2}{b^2} + \frac{a}{b} + 1)(\frac{a^2}{b^2} - \frac{a}{b} + 1)$

3. $(\frac{x^2}{y^2} + 1 + \frac{y^2}{x^2})(\frac{x^2}{y^2} - 1 + \frac{y^2}{x^2})$

4. $(p + 1)(p - 1)(p^2 + p + 1)(p^2 - p + 1)$

5. $(x + y - 2)(x - y - 8)$

6. $(ab + 10a + 3b + 30)(ab - 10a - 3b + 30)$

7. $(a + b + 1)(6a + 6b - 7)$

8. $(\frac{2x}{y} - \frac{5y}{x})(\frac{x}{y} + \frac{y}{x})$

2. H.C.F. and L.C.M. and simplification

1. Find the H.C.F. of $x^3 - y^3$, $x^6 - y^6$, $x^4 + x^2y^2 + y^4$

Solution:

Here, $x^3 - y^3$, $x^6 - y^6$, $x^4 + x^2y^2 + y^4$

The 1st expression $= x^3 - y^3 = (x - y)(x^2 + xy + y^2)$

The 2nd expression $= x^6 - y^6 = (x^3 + y^3)(x^3 - y^3)$

$$= (x + y)(x^2 - xy + y^2)(x - y)(x^2 + xy + y^2)$$

$$= (x + y)(x - y)(x^2 + xy + y^2)(x^2 - xy + y^2)$$

The 3rd expression $= x^4 + x^2y^2 + y^4$

$$= (x^2)^2 + (y^2)^2 + x^2y^2$$

$$= (x^2 + y^2)^2 - 2x^2y^2 + x^2y^2$$

$$= (x^2 + y^2)^2 - x^2y^2 = (x^2 + y^2)^2 - (xy)^2 = (x^2 + xy + y^2)(x^2 - xy + y^2)$$

$$\therefore \text{H.C.F.} = x^2 + xy + y^2$$

2. Simplify: $\frac{1}{(a-b)^2} - \frac{1}{a^2 - b^2}$

Solution:

$$\begin{aligned}\text{Here, } \frac{1}{(a-b)^2} - \frac{1}{a^2 - b^2} &= \frac{1}{(a-b)(a-b)} - \frac{1}{(a+b)(a-b)} \\ &= \frac{(a+b) - (a-b)}{(a-b)(a-b)(a+b)} = \frac{a+b-a+b}{(a-b)^2(a+b)} = \frac{2b}{(a+b)(a-b)^2}\end{aligned}$$

3. Simplify: $\frac{2y+5}{y^2+6y+9} - \frac{11}{y^2-9} - \frac{16y}{8y^2-24y}$

Solution:

$$\begin{aligned}\text{Here, } \frac{2y+5}{y^2+6y+9} - \frac{11}{y^2-9} - \frac{16y}{8y^2-24y} \\ &= \frac{2y+5}{(y+3)^2} - \frac{11}{(y+3)(y-3)} - \frac{16y}{8y(y-3)} \\ &= \frac{(2y+5)(y-3) + 11(y+3) - 2(y+3)^2}{(y+3)^2(y-3)} \\ &= \frac{2y^2 - 6y + 5y - 15 + 11y + 33 - 2(y^2 + 6y + 9)}{(y+3)^2(y-3)} \\ &= \frac{2y^2 + 10y + 18 - 2y^2 - 12y - 18}{(y+3)^2(y-3)} = \frac{-2y}{(y+3)^2(y-3)} = \frac{2y}{(3-y)(y+3)^2}\end{aligned}$$

4. Simplify: $\frac{a+2}{1+a+a^2} - \frac{a-2}{1-a+a^2} - \frac{2a^2}{1+a^2+a^4}$

Solution:

$$\begin{aligned}\text{Here, } \frac{a+2}{1+a+a^2} - \frac{a-2}{1-a+a^2} - \frac{2a^2}{1+a^2+a^4} \\ &= \frac{(a+2)(1-a+a^2) - (a-2)(1+a+a^2)}{(1+a+a^2)(1-a+a^2)} - \frac{2a^2}{(1+a^2)^2 - 2a^2 + a^2} \\ &= \frac{a - a^2 + a^3 + 2 - 2a + 2a^2 - a - a^2 - a^3 + 2 + 2a + 2a^2}{(1+a+a^2)(1-a+a^2)} - \frac{2a^2}{(1+a+a^2)(1-a+a^2)} \\ &= \frac{2a^2}{(1+a+a^2)(1-a+a^2)} - \frac{2a^2}{(1+a+a^2)(1-a+a^2)} = \frac{2a^2 - 2a^2}{(1+a+a^2)(1-a+a^2)} = 0\end{aligned}$$

Extra questions:

- Find the H.C.F. and L.C.M. of
 a) $a^2 - b^2$, $a^3 - b^3$ and $a^4 - b^4$ b) $a^2 + 2ab + b^2 - c^2$, $b^2 + 2bc + c^2 - a^2$, $c^2 + 2ca + a^2 - b^2$
 c) $x^3 + y^3$, $x^4 + x^2y^2 + y^4$ d) $x^2 - 4$, $x^3 + 8$, $x^2 + 5x + 6$
 e) $(a+b)^2 - 4ab$, $a^3 - b^3$, $a^2 + ab - 2b^2$

2. Simplify:

$$\begin{aligned}\text{a) } \frac{x-y}{xy} + \frac{y-z}{yz} + \frac{z-x}{zx} & \quad \text{b) } \frac{a^2-ab+b^2}{a-b} + \frac{a^2+ab+b^2}{a+b} \\ \text{c) } \frac{x}{xy-y^2} + \frac{y}{xy-x^2} & \quad \text{d) } \frac{a^2+b^2}{ab} - \frac{b^2}{a(a+b)} - \frac{a^2}{b(a+b)} \\ \text{e) } \frac{(a-b)^2 - c^2}{a^2 - (b+c)^2} + \frac{(b-c)^2 - a^2}{b^2 - (c+a)^2} + \frac{(c-a)^2 - b^2}{c^2 - (a+b)^2}\end{aligned}$$

Answers

- a) $(a-b)(a^4 - b^4)(a^2 + ab + b^2)$ b) $a + b + c$, $(a+b+c)(a+b-c)(b+c-a)(c+a-b)$
 c) $(x^2 - xy + y^2)$, $(x+y)(x^4 + x^2y^2 + y^4)$ d) $(x+2)$, $(x-2)(x+3)(x^3 + 8)$
 e) $(a-b)$, $(a-b)^2(a+2b)(a^2 + ab + b^2)$ 2. a) 0 b) $\frac{2a^3}{a^2 - b^2}$ c) $\frac{x+y}{xy}$ d) 1 e) 1